

HFLAD: AI-Driven Hierarchical Fusion Learning for Predictive Anomaly Detection in Consumer-Centric Systems

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Abstract—Audit logs, crucial for service auditing and security, capture user and system activities, which can be effectively represented as multivariate time series data to enable efficient log analysis, such as behavior analytics for anomaly detection. However, it is technically challenging to analyze such data due to its high dimensionality and strong time correlation. To address these challenges and enhance predictive security for consumer applications, we propose an AI-driven anomaly detection model named HFLAD. This model comprehensively considers the feature and time correlation of multivariate time series data. It employs causal and dilated convolutions for encoding to capture multi-scale temporal dependencies, and utilizes an HVAE (Hierarchical Variational Autoencoder) generator for information reconstruction. Then, it identifies anomalies by leveraging the difference between the reconstructed data and the original data. By leveraging hierarchical fusion learning, HFLAD proactively detects deviations in consumer-related system activities and audit logs, aligning with modern security demands in smart devices, Internet of Things (IoT), and online services. Experimental results demonstrate that HFLAD outperforms other existing techniques in identifying anomalies, thereby improving the effectiveness of predictive security measures for consumer systems.

Index Terms—Multivariate Time Series, Anomaly Detection, Feature Correlation, Time Correlation

I. INTRODUCTION

In consumer-facing systems such as smart appliances, wearables, and mobile services, audit logs and telemetry streams are increasingly used to track user-device interactions and operational status. These data, typically modeled as multivariate time series, form the foundation for AI-driven predictive

security. As consumer devices become more interconnected and autonomous, ensuring their behavioral reliability through predictive anomaly detection is essential [1]–[3].

Notably, consumer systems bear unique characteristics that make early anomaly detection far more critical than in traditional industrial or enterprise scenarios. For smart home ecosystems [4], for instance, a delayed response to anomalies in HVAC (heating, ventilation, and air conditioning) sensor data could lead to excessive energy consumption or equipment overheating[1]. In wearable devices, timely identification of irregularities in heart rate or blood glucose sensor streams is directly tied to user health, delays could result in missed warnings for potential medical emergencies, a risk emphasized in studies of healthcare wearables where real-time anomaly alerts improve clinical intervention timelines [5]. For mobile services, early detection of anomalous login patterns (e.g., unrecognized locations or unusual device types) can block account breaches before sensitive user data (such as payment information or personal messages) is compromised, a capability highlighted as critical for consumer privacy protection in mobile security frameworks [6].

Traditional approaches such as statistical methods (e.g., LOF and Gaussian mixture models) and machine learning techniques (e.g., clustering or density-based methods) have laid the foundation for anomaly detection [7]–[9]. However, these methods often fail to fully capture the intricate temporal and feature-based interdependencies of multivariate time series data [10], which are particularly common in telemetry streams generated by smart consumer devices, such as home sensors, wearables, and IoT services.

More recently, deep learning models like LSTM [11], VAE [12], and GAN [13] have demonstrated their ability to model complex patterns, but they typically focus on either temporal or feature-based attributes, leading to incomplete representations. This limitation is especially problematic for predictive security in consumer environments, where anomalies may manifest through subtle, multi-dimensional interactions, such as coordinated behavior deviations, delayed fault accumulation, or cross-device anomalies [14]. Hence, there is a pressing need for AI-driven models that can holistically integrate temporal dynamics and feature relationships to enable early and accurate identification of abnormal behavior in consumer-facing applications.

Existing anomaly detection models for multivariate time series primarily encode either feature correlations or temporal

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dynamics but often neglect their joint modeling, leading to the following limitations:

Isolated Modeling: Most approaches focus on feature correlations or temporal dependencies in isolation, which limits the ability to fully represent the interdependencies and complexity inherent in multivariate time series data [15], [16].

Single Time-Scale Modeling: Short time-span encodings often miss critical long-term dependencies, while long time-span encodings may overlook local patterns. This reliance on a single time scale prevents effective modeling of inter-scale temporal impacts [17].

To address these challenges, we propose HFLAD, a hierarchical fusion learning-based anomaly detection model that captures both temporal trends and latent feature dependencies. HFLAD leverages multiscale time encoders and a relational feature extractor to model normal system behavior and identifies anomalies based on reconstruction deviation. This design enables predictive detection of subtle behavioral shifts in consumer-centric systems before service disruption or security compromise occurs. The key contributions of this work include:

- (1) We propose a predictive anomaly detection framework (HFLAD) designed for multivariate time series data, particularly suited for consumer-centric systems such as smart home platforms and mobile services. It integrates temporal modeling and relational reasoning to address complex behavior monitoring needs.
- (2) We design a hierarchical fusion architecture that incorporates a multiscale time encoder and latent relational modeling via a structured recurrent neural network (SRNN), enabling proactive detection of subtle and evolving anomalies.
- (3) We implement and evaluate HFLAD on four real-world datasets, demonstrating its superior detection performance across multiple anomaly types and scenarios. The results highlight its effectiveness in early identification of risks in consumer-facing applications.

The rest of this paper is organized as follows: Section II reviews related work, Section III details the proposed HFLAD model, Section IV evaluates its performance, Section V presents a discussion on this work and Section VI concludes this article.

II. RELATED WORK

Anomaly detection for multivariate time series has evolved from statistical approaches to machine learning and deep learning techniques. A summary of their advantages and limitations is provided in Table I.

A. Statistic-based Methods

Statistical anomaly detection methods typically fit data to a predefined distribution and identify deviations as anomalies. Parametric approaches, such as Gaussian mixture models [18], assume that data follow fixed distributions, while non-parametric techniques, such as kernel density estimation [19], do not require prior distribution assumptions. Although statistical methods are interpretable and efficient for small datasets,

they often struggle with scalability and high-dimensional data complexity.

B. Machine Learning Methods

Machine learning techniques extend statistical methods by introducing more flexible and adaptive algorithms. Density-based methods, such as LOF [7], identify anomalies by measuring the density of local neighborhoods. Distance-based approaches [20] flag anomalies based on deviations from proximity thresholds. These methods can effectively detect outliers but are often sensitive to parameter tuning and computationally intensive for large-scale datasets.

C. Deep Learning Methods

Deep learning has driven significant advancements in anomaly detection, particularly for complex, high-dimensional multivariate time series. Prediction-based models, such as LSTM-NDT [11] and HilBERT [21], forecast time series data and use prediction errors to detect anomalies. Reconstruction-based methods, including LSTM-VAE [12] and DAGMM [8], learn latent representations to reconstruct input data, identifying anomalies as reconstruction errors. GAN-based models, such as MAD-GAN [22], use adversarial training to implicitly model normal data distributions and detect deviations [23].

While these approaches have achieved success, most focus on either temporal dependencies or feature correlations, limiting their ability to comprehensively model multivariate time series. Recent efforts, such as OmniAnomaly [13] and InterFusion [10], have attempted joint modeling of temporal and feature information. However, challenges remain in capturing inter-scale temporal patterns and complex inter-variable relationships.

In recent years, methods based on Transformers and related temporal modeling have significantly advanced robust anomaly detection [24], [25] for high-dimensional multivariate time series. Addressing the scarcity of anomalous samples, Anomaly Transformer [26] proposes the "adjacency concentration bias" hypothesis, computes association discrepancies via an anomaly-attention mechanism, and amplifies the discriminability between normal and anomalous samples, achieving state-of-the-art (SOTA) results across multiple scenarios. TimesNet [27] innovatively transforms 1D time series into 2D tensors, integrating 2D convolutions with adaptive Fourier transforms to capture multi-scale temporal patterns, demonstrating outstanding performance on non-stationary industrial data. Dual-TF [28] adopts a dual-branch Transformer architecture to process time-domain and frequency-domain data in parallel, enhancing detection accuracy for noisy industrial scenarios.

III. THE STRUCTURE OF HFLAD

Our anomaly detection model, HFLAD, reconstructs multivariate time series data to uncover abnormal patterns deviating from normal distributions. Unlike general data, multivariate time series possess two defining characteristics: temporal

TABLE I
ADVANTAGES AND LIMITATIONS OF DIFFERENT METHODS

Method	Advantages	Limitations
Statistic-based (Parametric)	Simple, reusable	Assumes fixed distribution
Statistic-based (Non-parametric)	Flexible, adaptive	Greater complexity
Density-based (ML)	Identifies low-density anomalies	Sensitive to density thresholds
Distance-based (ML)	Effective for local patterns	Parameter-dependent
Prediction-based (DL)	Good at time series forecasting	Relies on representation learning
Reconstruction-based (DL)	Extracts informative latent features	Complex latent space learning
GAN-based (DL)	Implicit modeling of normal data	Difficulty overcoming mode collapse

dependencies and feature correlations. Effective anomaly detection requires jointly modeling both attributes, which many existing methods fail to address.

As shown in Figure 1, the proposed model consists of three key modules:

- **Hierarchical Time Encoder:** Captures multi-scale temporal patterns using causal, dilated, and temporal convolutional networks.
- **Feature Encoder:** Extracts inter-variable relationships with dimensionality reduction.
- **HVAE-Based Generator:** Reconstructs the original input by combining multi-level latent representations.

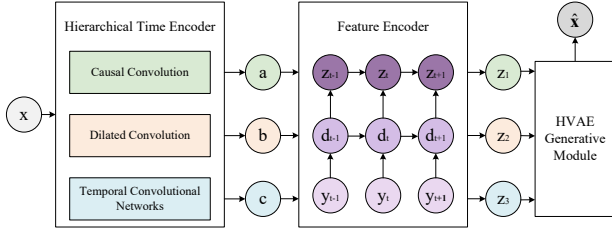


Fig. 1. Model Architecture

A. Hierarchical Time Encoder

The time encoding module processes input data $\mathbf{X} \in \mathbb{R}^{M \times T}$, where M represents the feature dimension and T denotes the time series length. This module integrates three techniques, causal convolution, dilated convolution, and temporal convolutional networks (TCN)-to capture both short- and long-term temporal patterns.

1) *Causal Convolution*: As shown in Figure 2, the causal convolutional network module can be divided into input, hidden, and output layers. The hidden layers, which is the key module and performs time series encoding. Causal convolution ensures the output at time t depends only on the current and preceding inputs, respecting temporal causality. Given input $\mathbf{x} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T\}$ over time T , the causal convolution computes the encoded output $\mathbf{a} = \{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_T\}$ with K hidden layers is computed as:

$$a_t = \sum_{k=1}^{K-1} h_k \mathbf{x}_{t-k} \quad (1)$$

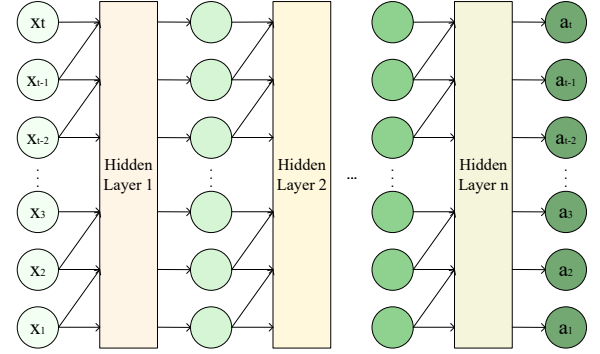


Fig. 2. Causal Convolution

where K is the kernel size and h_k denotes the filter coefficients. This operation efficiently captures local temporal dependencies and provides a foundation for hierarchical encoding. In Algorithm 1, $g_1(*)$ serves as the causal convolution module to generate the encoded sequence $\mathbf{a}$.

2) *Dilated Convolution*: To capture long-term dependencies, dilated convolution expands the receptive field by introducing a dilation rate l . As depicted in Figure 3, dilated convolution shares the causal convolution architecture but introduces dilation rate l within convolutional hidden layers. For input $\mathbf{x}$, the encoded output with K hidden layers is given by:

$$\mathbf{b}_t = \sum_{k=0}^{K-1} h_k \mathbf{x}_{t-kl}. \quad (2)$$

The receptive field span $(K-1)l+1$, allowing information from distant time points to be encoded efficiently. By adjusting l and K , the encoder dynamically balances between local and global information. In Algorithm 1, $g_2(*)$ serves as the dilated convolution module to generate the encoded sequence $\mathbf{b}$.

3) *Temporal Convolutional Network (TCN)*: TCNs integrate causal and dilated convolutions (DC) with additional enhancements, such as weight normalization (WN), dropout, and ReLU activation, ensuring stable training and better scalability. The TCN outputs $\mathbf{c}$ are computed hierarchically, combining information across multiple temporal scales:

$$\text{tmp}(x) = \text{Dropout}(\text{ReLU}(\text{WN}(\text{DC}(\mathbf{x})))) \quad (3)$$

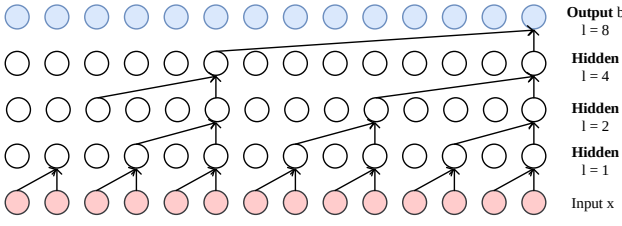
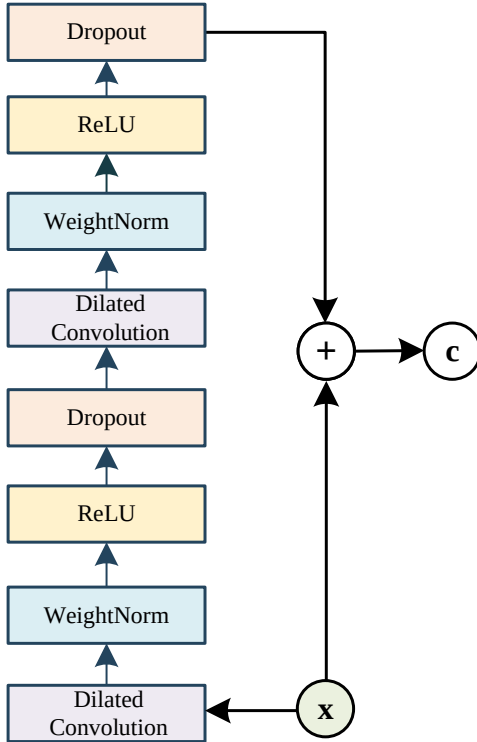


Fig. 3. Dilated Convolution

$$\mathbf{c}_t = \sum_{k=0}^{K-1} w_k (\text{tmp}(\text{tmp}(\mathbf{x})))_{t-lk} + \mathbf{x}_t \quad (4)$$

where w_k are learnable weights. As shown in Figure 4, TCN effectively captures both local and long-term dependencies, making it a critical component of the hierarchical time encoder. In Algorithm 1, $g_3(\cdot)$ serves as the temporal convolution network to generate the encoded sequence $\mathbf{c}$.

Fig. 4. Temporal Convolutional Network. It captures local and long term dependencies, serving as a key part of the hierarchical time encoder to generate $\mathbf{c}$.

B. Feature Encoder

After the Hierarchical Time Encoder generates multi-scale temporal feature sequences $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ via causal convolution, dilated convolution, and TCN, these sequences are fed into the Feature Encoder as independent inputs $\mathbf{y}$ (e.g., $\mathbf{y} = \mathbf{a} = \{a_1, a_2, \dots, a_T\}$). The core goal of this module is to map these temporal features to latent variables z_1, z_2 , and z_3 (via the mapping function $g_4(\cdot)$) while preserving inter-variable

relationships and reducing dimensionality. To achieve this, we adopt an adapted State-Space Recurrent Neural Network (SRNN) architecture that integrates state space models with recurrent neural networks, enabling joint encoding of feature and temporal dependencies.

At each time step t , the SRNN computes the hidden state d_t and random latent variable z_t by balancing the update of new information and retention of historical states. The hidden state d_t is defined as:

$$\mathbf{d}_t = f(\mathbf{u}_t, \mathbf{d}_{t-1}) + (1 - \mathbf{u}_t) \mathbf{d}'_t, \quad (5)$$

where u_t is the update gate that controls how much new information is integrated, d'_t is the temporary state capturing current input features, and f is a learned transition function.

To compute u_t and d'_t , we introduce the reset parameter r_t that regulates the influence of the previous hidden state d_{t-1} . The update gate u_t and reset parameter r_t are calculated using the sigmoid activation function σ (which outputs values between 0 and 1) as:

$$\mathbf{u}_t = \sigma(\mathbf{A}_n \mathbf{y}_t + \mathbf{C}_n \mathbf{d}_{t-1} + \text{bias}_n), \quad (6)$$

$$\mathbf{r}_t = \sigma(\mathbf{A}_r \mathbf{y}_t + \mathbf{C}_r \mathbf{d}_{t-1} + \text{bias}_r). \quad (7)$$

Here, $\mathbf{A}_n$, $\mathbf{C}_n$ (and analogous matrices for r_t) are learnable parameters, and bias_n (and bias_r) are bias matrices that adjust the linear transformations of the input $\mathbf{y}_t$, and previous hidden state $\mathbf{d}_{t-1}$. The temporary state $\mathbf{d}'_t$ is then computed using the tanh activation function (which constrains values between -1 and 1) to introduce non-linearity, with the reset parameter r_t modulating the contribution of $\mathbf{d}_{t-1}$ via element-wise multiplication ($\odot$):

$$\mathbf{d}'_t = \tanh(\mathbf{A}_d \mathbf{y}_t + \mathbf{r}_t \odot (\mathbf{C}_d \mathbf{d}_{t-1}) + \text{bias}_d), \quad (8)$$

To model the statistical distribution of latent variables, we assume the joint distribution of z and d follows a sequential probabilistic structure, where each time step's variables depend on the previous step's state:

$$\begin{aligned} p_{\theta}(\mathbf{z}, \mathbf{d} | \mathbf{y}, \mathbf{z}_0, \mathbf{d}_0) &= p_{\theta_z}(\mathbf{z} | \mathbf{d}, \mathbf{z}_0) p_{\theta_d}(\mathbf{d} | \mathbf{y}, \mathbf{d}_0) \\ &= \prod_{t=1}^T p_{\theta_z}(\mathbf{z}_t | \mathbf{z}_{t-1}, \mathbf{d}_t) p_{\theta_d}(\mathbf{d}_t | \mathbf{d}_{t-1}, \mathbf{y}_t) \end{aligned} \quad (9)$$

with $\mathbf{z}_0$ and $\mathbf{d}_0$ initialized as zero vectors to provide a starting point for the sequence. We further specify that the conditional probability of z_t given z_{t-1} and d_t follows a Gaussian distribution, as Gaussian distributions effectively model continuous latent spaces and enable efficient learning:

$$p_{\theta_z}(\mathbf{z}_t | \mathbf{z}_{t-1}, \mathbf{d}_t) = \mathcal{N}(\mathbf{z}_t, \mu_t, \sigma_t^2 \mathbf{I}), \quad (10)$$

where μ_t (mean) and σ_t^2 (variance) are dynamically updated by two feedforward neural networks NN_1 and NN_2 that take $\mathbf{z}_{t-1}$ and $\mathbf{d}_t$ as inputs:

$$\mu_t = NN_1(\mathbf{z}_{t-1}, \mathbf{d}_t), \quad \sigma_t^2 = NN_2(\mathbf{z}_{t-1}, \mathbf{d}_t). \quad (11)$$

To optimize the model parameters $\theta = \{\theta_z, \theta_d\}$, we maximize the log-likelihood of the latent variables given the input

data, ensuring the model learns meaningful latent representations:

$$\tau(\theta) = \sum_m \log p_\theta(\mathbf{z}^m | \mathbf{y}^m, \mathbf{z}_0^m, \mathbf{d}_0^m), \quad (12)$$

where m indexes individual data points. For further refinement of hidden state dynamics, we simplify the transition function f to f_{θ_d} (parameterized by θ_d):

$$\mathbf{d}_t = f_{\theta_d}(\mathbf{y}_t, \mathbf{d}_{t-1}), \quad (13)$$

and define a regularization loss on the z_t using a distance function $\rho(\cdot)$ to prevent overfitting and stabilize training:

$$\mathcal{L}_{reg} = \sum_{t=1}^T \rho(z_t - f_{\theta_z}(z_{t-1}, d_t)), \quad (14)$$

C. HVAE-Based Generator

The Hierarchical Time Encoder and Feature Encoder collectively produce three latent representations z_1 , z_2 , and z_3 , each capturing distinct multi-scale temporal and feature dependencies. The HVAE-Based Generator's role is to fuse these hierarchical latent variables and reconstruct the original input $\mathbf{x}$ into $\hat{\mathbf{x}}$, following the pipeline:

$$\mathbf{x} \rightarrow \{z_1, z_2, z_3\}, \quad \{z_1, z_2, z_3\} \rightarrow \hat{\mathbf{x}}.$$

The quality of this reconstruction directly determines anomaly detection performance—anomalous data (unseen during training) will have larger discrepancies between $\mathbf{x}$ and $\hat{\mathbf{x}}$ than normal data. To achieve effective fusion and reconstruction, we employ a HVAE [29], which organizes latent variables into a hierarchical structure to model the intrinsic distribution of raw data. The key insight is to impose a probabilistic chain structure on the latent variables, where higher-level latent variables (capturing complex, global patterns) guide the distribution of lower-level variables (capturing fine-grained, local features). This hierarchical dependency is formalized as:

$$p_\theta(z_1, z_2) = p_\theta(z_1 | z_2) p_\theta(z_2). \quad (15)$$

Extending this to all three latent variables, the joint probability of the input $\mathbf{x}$ and latent variables z_1, z_2, z_3 is decomposed into a product of conditional probabilities, reflecting the hierarchical fusion process:

$$p_\theta(\mathbf{x}, z_1, z_2, z_3) = p_\theta(\mathbf{x} | z_1, z_2, z_3) p_\theta(z_1 | z_2, z_3) p_\theta(z_2 | z_3) p_\theta(z_3). \quad (16)$$

Based on the joint probability in Eq. (16), the marginal log-likelihood of the input x can be lower-bounded by the evidence lower bound :

$$\log p_\theta(x) \geq \mathbb{E}_{q_\phi(z_1, z_2, z_3 | x)} \left[\log \frac{p_\theta(x, z_1, z_2, z_3)}{q_\phi(z_1, z_2, z_3 | x)} \right]. \quad (17)$$

This lower bound provides a tractable objective for optimizing the HVAE-based generator.

The HVAE's architecture (illustrated in Fig.5) enables chain conduction between latent variables: z_3 (highest-level, complex features) influences z_2 , which in turn guides z_1 (lowest-level, fine-grained features). This structure ensures that the

fused latent representation comprehensively encodes all critical dependencies, enabling precise reconstruction of $\hat{x}$ for normal data and noticeable deviations for anomalies.

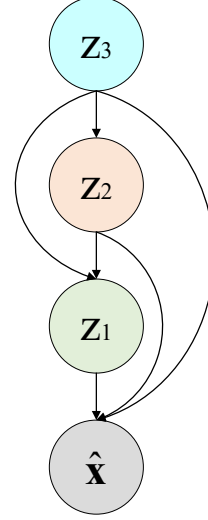


Fig. 5. Hierarchical Variational Autoencoder (HVAE) Generative Module Architecture. Hierarchical latent variables (z_1, z_2, z_3) interaction for precise $\hat{\mathbf{x}}$ reconstruction.

D. Model Training Algorithm

To optimize the model parameters and align the distributions of reconstructed data $\hat{\mathbf{x}}$ and original data $\mathbf{x}$, we design a loss function Γ_{loss} based on reconstruction accuracy, KL divergence and latent regularization:

$$\begin{aligned} \Gamma_{\text{loss}} = & -\mathbb{E}_{q_\phi(z_1, z_2, z_3 | \mathbf{x})} [\log p_\theta(\mathbf{x} | z_1, z_2, z_3)] \\ & + D_{\text{KL}}(q_\phi(z_1, z_2, z_3 | \mathbf{x}) \parallel p_\theta(z_1, z_2, z_3)) \\ & + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}}. \end{aligned} \quad (18)$$

The training algorithm alternates between encoding time and feature representations and reconstructing $\hat{\mathbf{x}}$ via HVAE. Parameters θ and ϕ are updated using stochastic gradient descent to minimize Γ_{loss} . The whole training process is detailed in Algorithm 1.

E. Abnormal Score

During training, reconstruction deviations guide parameter updates. For anomaly detection, we quantify abnormalities using a reconstruction-based anomaly score. Poor reconstruction on anomalous data results in higher scores, indicating deviations from normal patterns.

The anomaly score is defined as:

$$\begin{aligned} & -E_{q_\phi(z_1, z_2, z_3 | \mathbf{x})} [\log p_\theta(\mathbf{x} | z_1, z_2, z_3)] \\ & = -\frac{1}{M} \sum_{i=1}^M [\log p_\theta(\mathbf{x} | z_1^i, z_2^i, z_3^i)] \end{aligned} \quad (19)$$

Data points with scores exceeding a threshold λ are classified as anomalous, as they represent patterns unseen during training. This score effectively quantifies model uncertainty and highlights deviations in new data.

Algorithm 1 Training Algorithm

Input: Datasets $\mathbf{x}$
Output: Encoding distribution $q_\varphi(\mathbf{z}|\mathbf{x})$, reconstruction distribution $p_\theta(\hat{\mathbf{x}})$, parameter spaces θ and φ

Initialize φ, θ **for** $e \leftarrow 0$ **to** n **do**
 // n is number of iterations
 // time encoding stage
 $\mathbf{a} \leftarrow g_1(\mathbf{x})$; // g_1 : causal convolution
 $\mathbf{b} \leftarrow g_2(\mathbf{x})$; // g_2 : dilated convolution
 $\mathbf{c} \leftarrow g_3(\mathbf{x})$; // g_3 : TCN
 // feature encoding stage
 $\mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3 \leftarrow g_4(\mathbf{a}, \mathbf{b}, \mathbf{c})$; // g_4 : feature encoder
 // reconstruction stage
 $\hat{\mathbf{x}} \leftarrow h(\mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3)$; // h : HVAE
 $F_{\text{loss}} \leftarrow$ Calculate loss as Eq. 18 Update φ, θ via stochastic gradient descent
end
return θ, φ

IV. PERFORMANCE EVALUATION

This section evaluates HFLAD's performance on multivariate time series datasets, including comparisons with state-of-the-art models, analysis of detection capabilities on different anomaly types, complexity evaluations, and ablation studies. The source code of our experiment is available on GitHub (<https://github.com/1500848347-lgtm/HFLADmain>), where follow-up research updates will also be explained.

A. Datasets and Experimental Setup

HFLAD is evaluated on four public multivariate time series datasets:

- **SWaT** [30]: Water recycling system with 7 days of normal operation and 4 days of attack, which can simulate changes in multivariate time series under scenarios such as equipment anomalies and malicious attacks in industrial environments. It is a classic benchmark dataset for anomaly detection in the industrial control field.
- **KDD-Cup99** [31]: Constructed based on 9 weeks of network connection data collected by simulating the U.S. Air Force local area network environment, including 7 weeks of training data and 2 weeks of test data. Anomaly types are subdivided into 4 major categories (DOS, R2L, U2R, PROBING) with a total of 39 types, which can effectively verify the model's ability to identify network attack behaviors.
- **ASD** [10]: Constructed based on 45 days of operation monitoring data from 12 servers. Anomaly types mainly include server performance degradation, resource overload, and service response delay, which are consistent with the actual operation and maintenance monitoring scenarios of server clusters in data centers.

Key statistics of the datasets, including the sizes of training/test set, number of dimensions (NoD), and anomaly rates (ANR), are summarized in Table II. Each dataset contains distinct anomaly types, such as cyberattacks, sensor malfunctions,

intrusion attempts, and system degradations, reflecting real-world challenges in service auditing and security.

TABLE II
THE DETAIL OF DATASETS

Dataset	Training set	Test set	NoD	ANR(%)
SWaT	475200	449919	51	12.13
KDD	56139	24602	34	5.69
ASD	102331	51840	19	4.61

Performance metrics include the following. Here, TP, FP, FN, and TN denote true positives, false positives, false negatives, and true negatives, respectively:

- Precision (P): Fraction of predicted positives that are true positives, $P = \frac{TP}{TP+FP}$.
- Recall (R): Fraction of actual positives correctly predicted, $R = \frac{TP}{TP+FN}$.
- F1 Score: Harmonic mean of precision and recall, $F1 = 2 \frac{PR}{P+R}$.
- AUC: Evaluates model discrimination ability, where higher values indicate better performance with fewer false positives.

B. Comparison with State-of-the-Art Models

HFLAD is compared with the following benchmarks:

- **MSCRED** [32]: Uses ConvLSTM networks with attention to model inter-variable correlations.
- **MAD-GAN** [22]: Leverages LSTM-based GANs for anomaly detection.
- **InterFusion** [10]: Fuses temporal modeling and ensemble learning.
- **TFAD** [33]: Utilizes time-frequency analysis for feature extraction.

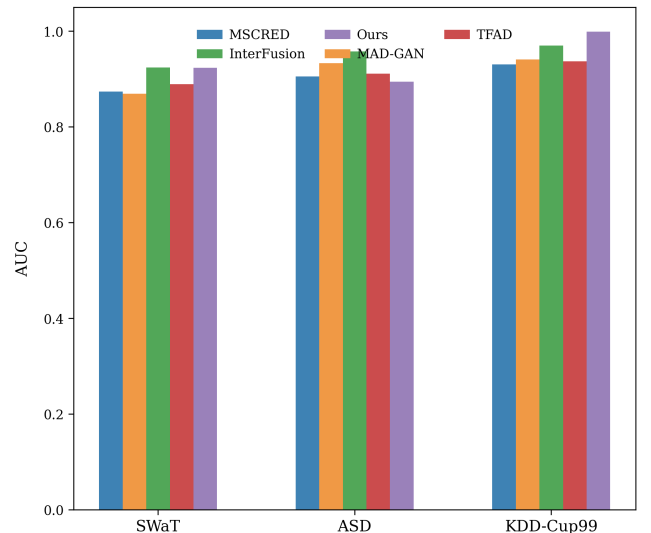


Fig. 6. The AUC values of models

TABLE III
EXPERIMENTAL RESULTS OF MODELS ACROSS DATASETS

Method	SWaT			ASD			KDD-Cup99		
	P	R	F1	P	R	F1	P	R	F1
MSCRED	0.9072	0.8425	0.8737	0.9152	0.8956	0.9053	0.9345	0.9256	0.9301
MAD-GAN	0.8526	0.8859	0.8689	0.9286	0.9377	0.9331	0.9255	0.9564	0.9407
InterFusion	0.9215	0.9266	0.9241	0.9542	0.9613	0.9577	0.9651	0.9743	0.9697
TFAD	0.9155	0.8639	0.8889	0.9015	0.9213	0.9113	0.9311	0.9425	0.9368
Ours	0.9677	0.9273	0.9471	0.9407	0.9811	0.9778	0.9732	0.9600	0.9665

Performance Results:

- SWaT: HFLAD improves InterFusion’s F1 score by 2.3%.
- ASD: Gains of 2.01% in F1 score.
- KDD-Cup99: HFLAD exceeds InterFusion by 1.66% in F1 and 2% in AUC.

Detailed Analysis:

- MSCRED: Focuses solely on temporal patterns, neglecting inter-variable dependencies, resulting in suboptimal performance.
- MAD-GAN: Excels at capturing sequence correlations but overlooks time-series dynamics.
- InterFusion: Combines temporal and feature modeling but lacks hierarchical depth, limiting its capacity to capture multi-scale dependencies.
- TFAD: Emphasizes time-frequency analysis, suitable for specific types of anomalies but less robust for diverse patterns.

HFLAD Advantage: The hierarchical time encoder and feature modeling in HFLAD provide comprehensive anomaly characterization, resulting in robust performance across datasets.

C. Detection of Different Anomaly Types

The ASD dataset includes three types of anomalies:

- Time Trend Anomalies: Deviations from historical patterns.
- Sequence Correlation Anomalies: Changes in inter-variable relationships.
- Bimodal Anomalies: A combination of trend and correlation deviations.

Figure 7 compares detection performance on these anomaly types, and visualizes the results.

Performance Analysis:

- MSCRED and TFAD: Detect time trend anomalies effectively due to their emphasis on temporal modeling.
- MAD-GAN: Performs better on sequence correlation anomalies by leveraging inter-variable dependencies.
- InterFusion: Fuses temporal and feature information, outperforming MSCRED, TFAD, and MAD-GAN.
- HFLAD: Surpasses some benchmarks, demonstrating its ability to detect diverse anomaly types through hierarchical temporal modeling and feature encoding.

These results highlight HFLAD’s capability to model both local and global dependencies, ensuring accurate detection of complex anomaly patterns.

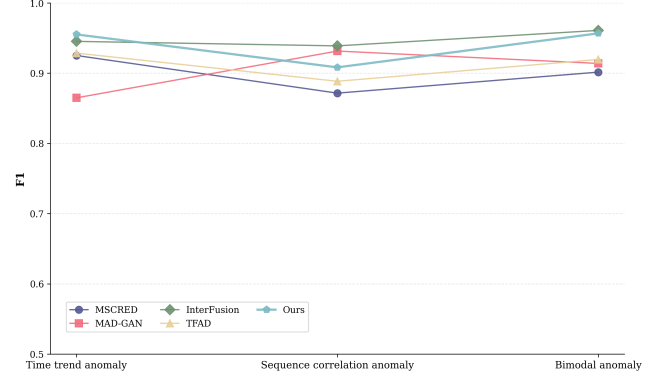


Fig. 7. F1 score of model in three abnormal modes of ASD dataset

D. Time and Space Complexity Analysis

Time Complexity: The time complexity of HFLAD is determined by the operations in hierarchical time encoding, feature encoding, and reconstruction with HVAE:

- Hierarchical Time Encoding: For n time-series points and t time scales, assuming linear operations at each time scale, the time complexity is $\mathcal{O}(nt)$.
- Feature Encoding: Assuming a linear transformation of the d -dimensional data, the time complexity is $\mathcal{O}(nd)$.
- Reconstruction (HVAE): If the HVAE model has l layers with a maximum of m nodes per layer, the time complexity for one data point is $\mathcal{O}(lmn)$. For n data points, the total time complexity is $\mathcal{O}(nlmn)$.

Thus, the overall time complexity of HFLAD is $\mathcal{O}(nt + nd + nlmn)$.

Space Complexity: The space complexity of HFLAD includes the storage for the time-series data, encoded features, and HVAE model parameters:

- Time-series Data: For n data points, each with d dimensions, the storage requirement is $\mathcal{O}(nd)$.
- Encoded Features: If the encoded feature space has p dimensions, the space required for storing the encoded features is $\mathcal{O}(np)$.
- HVAE Model Parameters: For a model with l layers and m nodes per layer, the storage required is $\mathcal{O}(lm)$.

Therefore, the overall space complexity is $\mathcal{O}(nd + np + lm)$.

E. Ablation Studies

To evaluate the contributions of HFLAD’s components, we conduct ablation experiments with the following variants:

TABLE IV
RESULTS OF ABLATION EXPERIMENT

Method	SWaT			ASD			KDD-Cup99		
	P	R	F1	P	R	F1	P	R	F1
A	0.8776	0.8952	0.8863	0.9236	0.9256	0.9246	0.9365	0.9255	0.9309
B	0.9012	0.8853	0.8932	0.9053	0.8999	0.9026	0.9018	0.9366	0.9189
C	0.9133	0.8746	0.8935	0.9325	0.9466	0.9395	0.9542	0.9433	0.9487
Ours	0.9677	0.9273	0.9471	0.9190	0.9636	0.9407	0.9732	0.9600	0.9665

- Scheme A: Replace the time encoder with a simple LSTM.
- Scheme B: Remove the feature encoder entirely.
- Scheme C: Replace the HVAE generator with a basic decoder.

Results and Analysis: By selectively replacing or removing key modules, we can quantify each component’s contribution to anomaly detection performance. Table IV and Figure 8 illustrate the impact of these modifications:

- Scheme A: HFLAD (Ours) outperforms Scheme A on all datasets, with F1-score improvements of 0.0608 (SWaT), 0.0161 (ASD), and 0.0356 (KDD-Cup99). Scheme A’s LSTM time encoder fails to capture hierarchical multi-scale temporal dependencies (e.g., simultaneous fine-grained pressure surges, flow rate fluctuations, and coarse-grained temperature in SWaT), reducing anomaly detection accuracy, while Ours’ hierarchical time encoder models multi-scale dependencies in parallel.
- Scheme B: HFLAD outperforms Scheme B on all datasets, with F1-score improvements of 0.0539 (SWaT), 0.0381 (ASD), and 0.0476 (KDD-Cup99). When Scheme B completely removes the feature encoder, the model fails to capture multivariate collaborative anomaly patterns due to the lack of inter-variable relationship modeling. In ASD, abnormal correlations between CPU usage and memory utilization cannot be effectively identified, while the feature encoder of the HFLAD explicitly learns inter-variable associations.
- Scheme C: HFLAD achieves comparable performance to Scheme C on ASD and outperforms it on the other datasets, with F1-score improvements of 0.0536 (SWaT), 0.0012 (ASD), and 0.0178 (KDD-Cup99). When Scheme C replaces the HVAE generator with a basic decoder, its simplistic reconstruction capability fails to capture complex data patterns, limiting anomaly detection performance. Taking KDD-Cup99 network traffic data as an example, complex attack traffic patterns cannot be accurately reconstructed by the basic decoder, while the HVAE generator of HFLAD can learn complex data distributions.

Summary: These findings validate the necessity of HFLAD’s components. The hierarchical temporal modeling, feature encoding, and HVAE generator synergistically enhance anomaly detection in multivariate time series.

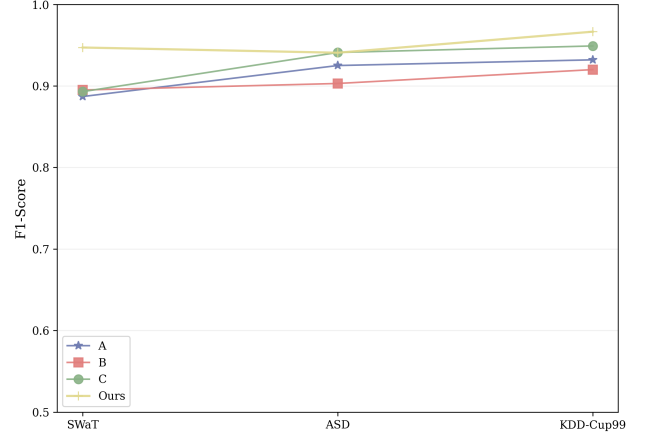


Fig. 8. F1 Score of model ablation experiment

V. DISCUSSION

This study focuses on consumer-centric systems (e.g., smart homes, wearables, IoT devices), where user interaction logs, sensor readings, and device operational metrics often contain sensitive information. Therefore, from the perspective of privacy preservation, this paper discusses the applicability of the HFLAD model in real-world scenarios to address the gap. To address this, HFLAD can be adapted to federated learning (FL) and differential privacy (DP) as follows:

For FL, a two-tier decentralized framework aligns with HFLAD’s modular design: resource-constrained consumer devices train lightweight local sub-models (e.g., simplified TCNs in the Hierarchical Time Encoder, compact HVAE decoders) using on-device data, edge servers aggregate local parameter updates via weighted federated averaging without centralizing raw data, and only high-confidence anomaly flags are shared for verification.

For DP, Gaussian noise is added to aggregated FL parameters during server-side aggregation, and Laplacian noise is injected into local anomaly scores to meet (ϵ, δ) -DP guarantees, with noise scales calibrated to balance privacy and detection performance.

These adaptations, combined with parameter pruning to reduce on-device overhead, enable HFLAD to retain its multi-scale temporal modeling and hierarchical reconstruction strengths while complying with regulations and protecting consumer privacy, making it viable for real-world deployment in smart homes, wearables, and IoT services.

VI. CONCLUSION

In this paper, we proposed HFLAD, a hierarchical fusion learning-based framework for predictive anomaly detection in multivariate time series. The model integrates multiscale temporal encoding, latent dependency reasoning, and a hierarchical variational reconstruction module to detect abnormal system behaviors. Through extensive experiments on benchmark datasets, HFLAD demonstrated robust performance across various anomaly types and datasets. By capturing subtle deviations in time-evolving consumer behavior logs, HFLAD addresses the growing need for predictive AI security in smart homes, wearable devices, and mobile services. Its modular design and general applicability make it suitable for deployment in diverse consumer-oriented scenarios. Future work will focus on real-time adaptation and edge deployment for resource-constrained consumer platforms, along with privacy-preserving learning extensions to ensure secure and scalable integration.

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